01 · VISION

See the shock *before it hits.*

See the supply shock coming; borrow the capacity to survive it.

PRODUCT

ShockTwin

Detect · Simulate · Respond · supplied brief

TEAM · RV UNIVERSITY

Francis Reuben R · Avaneesh Affan · Vivek

02 · THE COST

OVERALL CHIP SHORTAGE · 2021

One fire.
$210 billion lost.

The total reflects the overall chip shortage in 2021, not the loss attributable to one fire.

VEHICLES NOT BUILT

7.7M

AlixPartners, Sep 2021

AUTOMOTIVE MCU SHARE

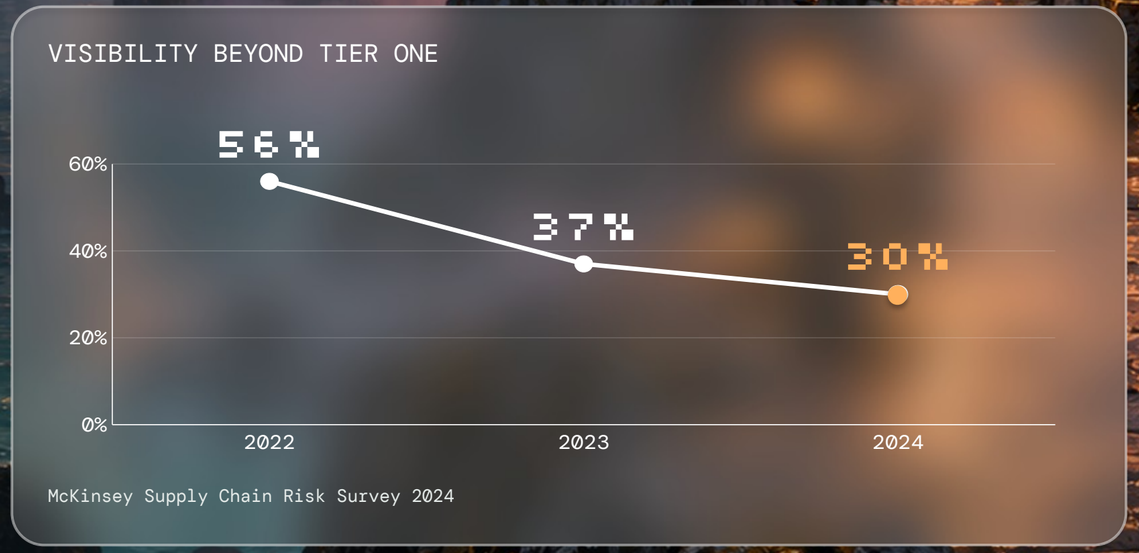
~30%

Reuters, Mar 2021; Renesas

03 · VISIBILITY

Most firms can't see *past tier 1.*

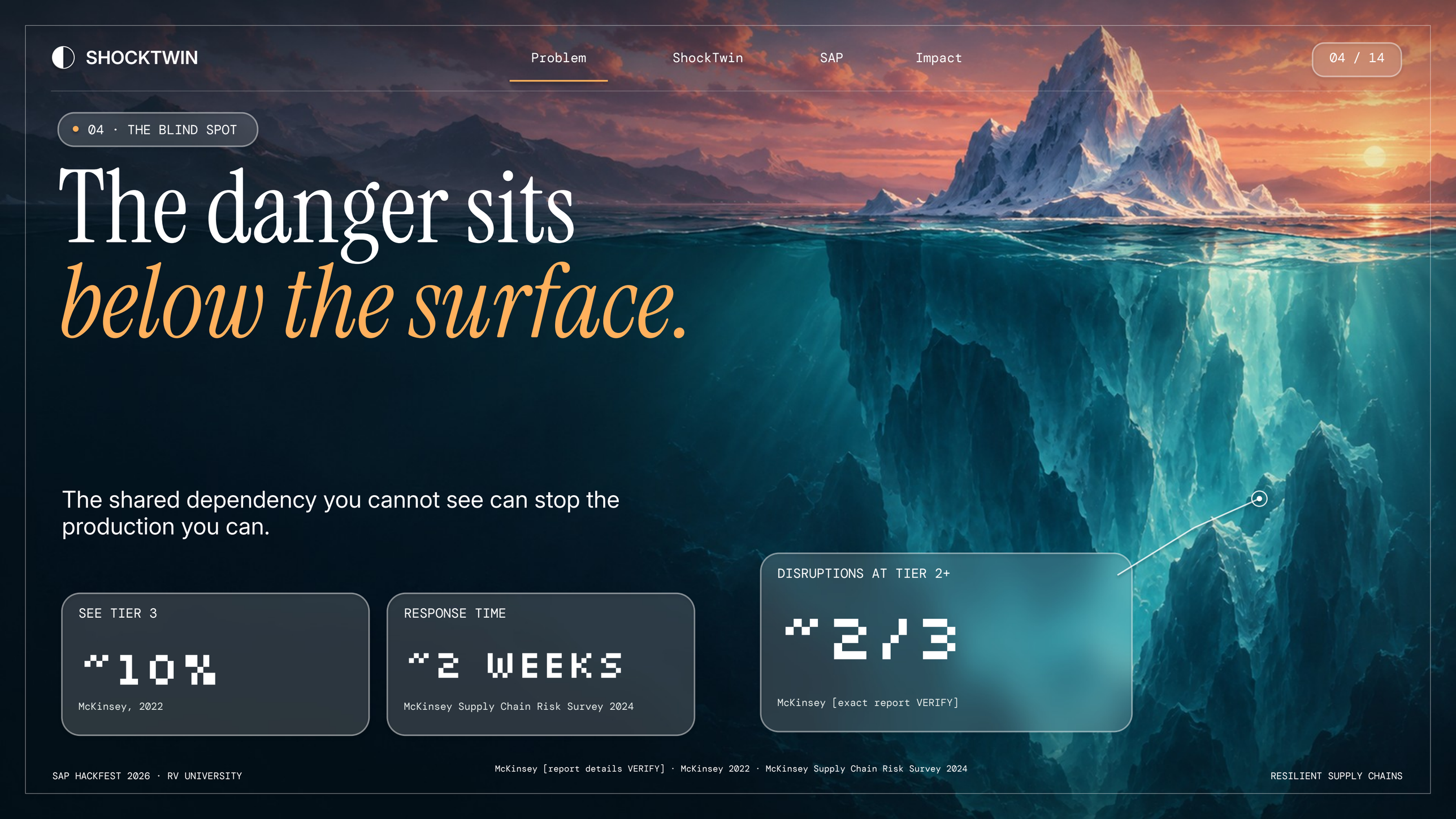
Beyond-tier-one visibility is shrinking when upstream exposure matters most.



McKinsey Supply Chain Risk Survey 2024

04 · THE BLIND SPOT

The danger sits *below the surface.*



The shared dependency you cannot see can stop the production you can.

SEE TIER 3

~10%

McKinsey, 2022

RESPONSE TIME

~2 WEEKS

McKinsey Supply Chain Risk Survey 2024

DISRUPTIONS AT TIER 2+

~2/3

McKinsey [exact report VERIFY]

05 · ROUTE RISK

The Red Sea closed. *India paid.*

Disrupted shipping routes transmitted cost pressure to Indian supply chains; the sea was not literally closed.



SUEZ TRAFFIC

-57%

UNCTAD; from peak to Oct 2024

SHANGHAI-EUROPE SPOT

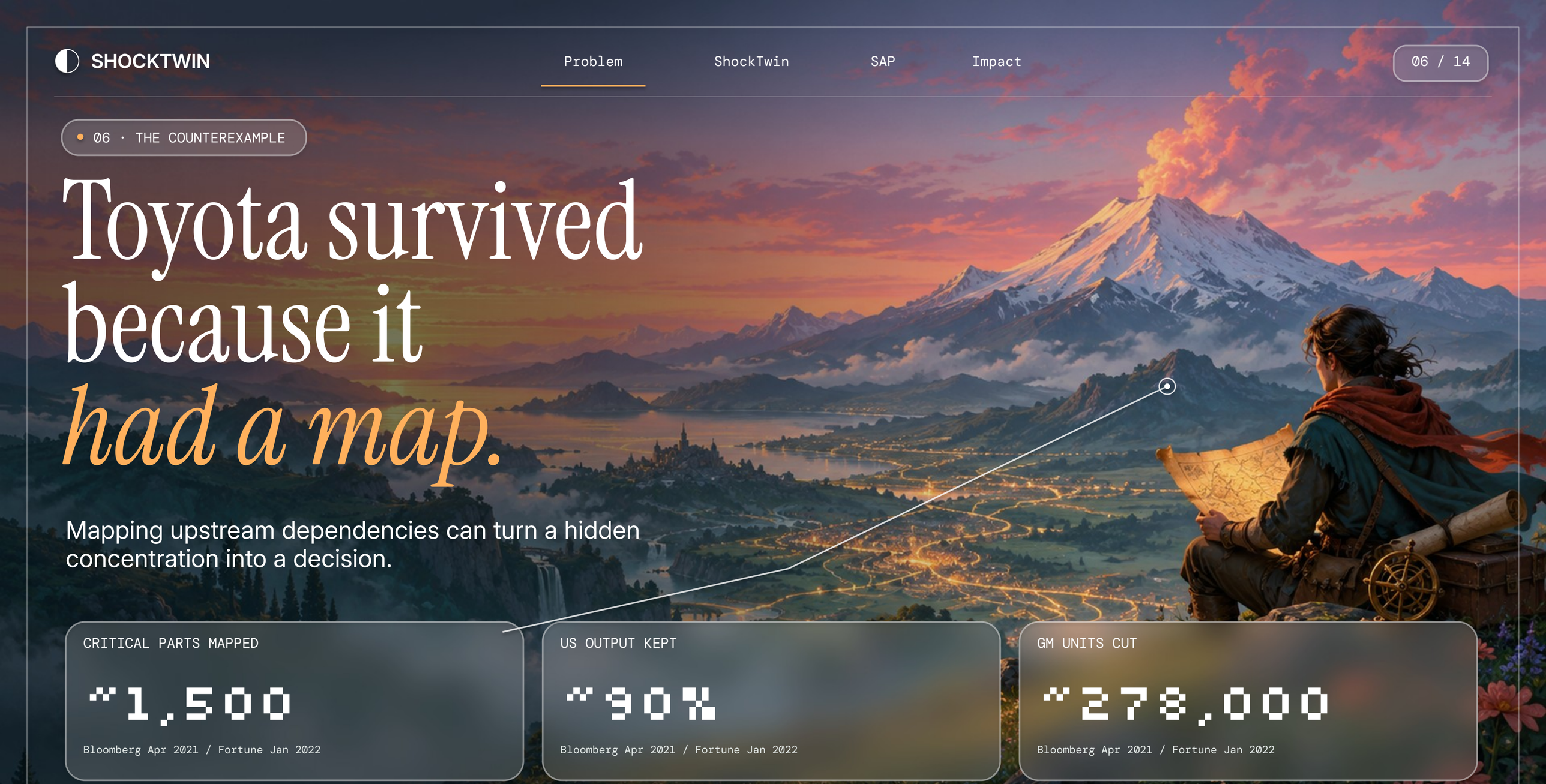
+256%

UNCTAD, 2024

06 · THE COUNTEREXAMPLE

Toyota survived because it *had a map.*

Mapping upstream dependencies can turn a hidden concentration into a decision.



CRITICAL PARTS MAPPED

~1,500

Bloomberg Apr 2021 / Fortune Jan 2022

US OUTPUT KEPT

~90%

Bloomberg Apr 2021 / Fortune Jan 2022

GM UNITS CUT

~278,000

Bloomberg Apr 2021 / Fortune Jan 2022

07 · DETECT · SIMULATE · RESPOND

Meet *ShockTwin.*

One connected loop: Detect, Simulate, Respond.

01

Detect

Supplied product loop

02

Simulate

Supplied product loop

03

Respond

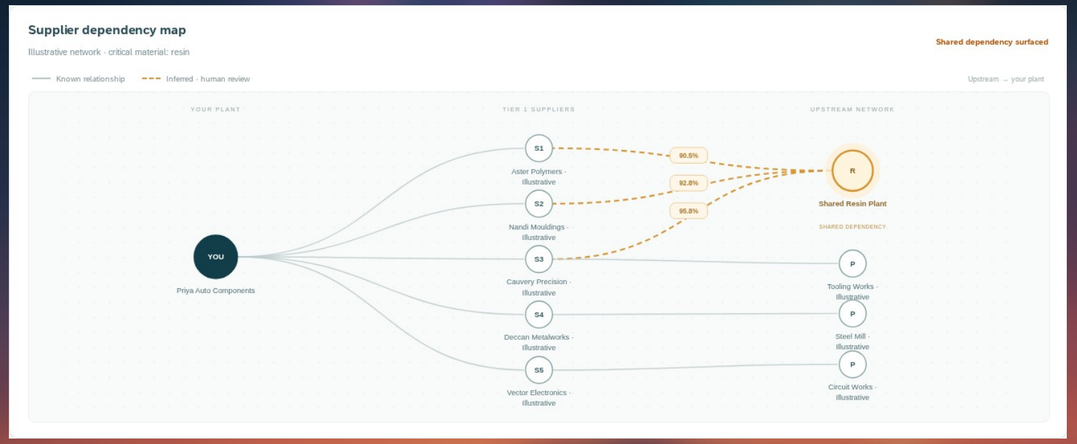
Supplied product loop

08 · 01 · DETECT

Find the suppliers *you never knew.*

Illustrative: three immediate suppliers share one hidden resin plant.

DETECT · REAL APP · ILLUSTRATIVE



Illustrative local app · actual browser capture

IMMEDIATE SUPPLIERS

5

Illustrative seed

SHARED UPSTREAM PLANT

1

Illustrative seed

DEPENDENT SUPPLIERS

3 OF 5

Illustrative seed

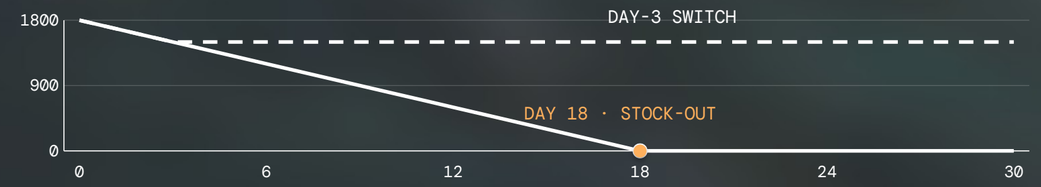
09 · 02 · SIMULATE

War-game the storm *before it lands.*

Illustrative: a day-three replacement switch avoids a line stop through the displayed simulation horizon.

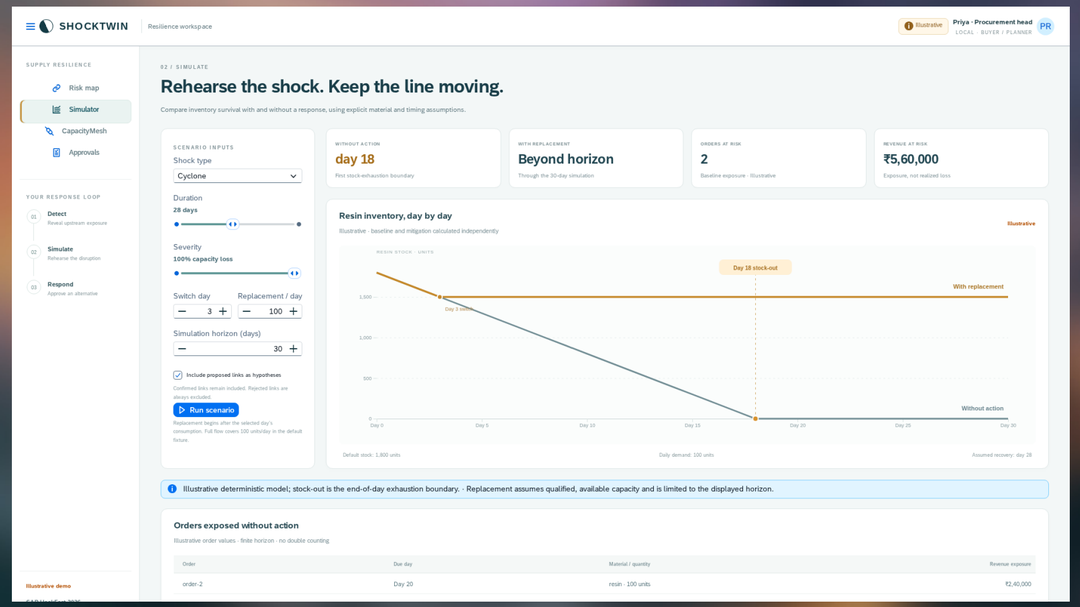
RESIN STOCK LEFT · ILLUSTRATIVE

DAY 18



Real local API · 100 units/day · 30-day horizon

SIMULATE · REAL APP · ILLUSTRATIVE



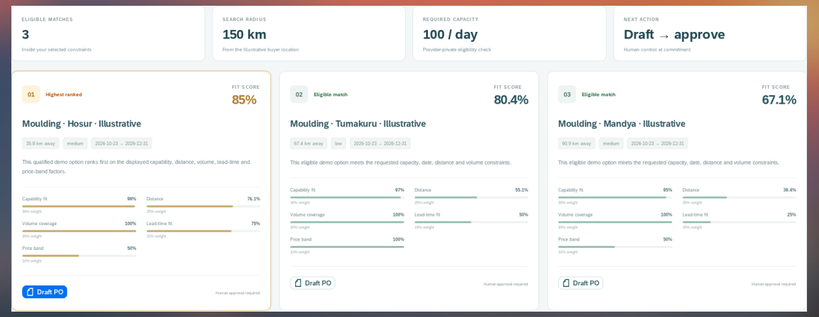
Illustrative local app · actual browser capture

10 · 03 · RESPOND · CAPACITYMESH

Borrow capacity *in minutes,* not weeks.

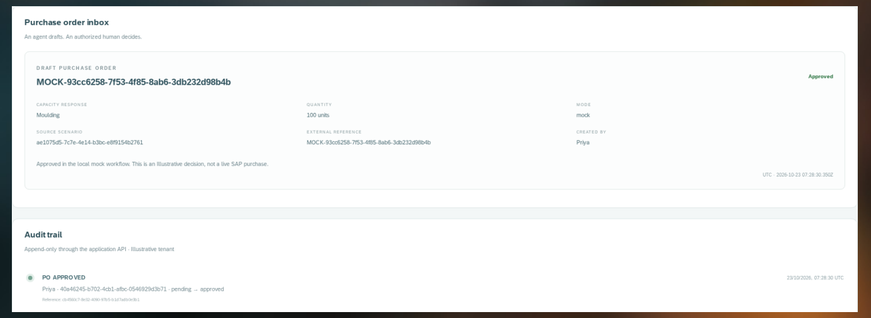
CapacityMesh turns a qualified match into a human-approved draft; minutes describes the demo workflow, not physical activation.

RANKED CAPACITY · ILLUSTRATIVE



Illustrative local app · actual browser capture

HUMAN APPROVED · MOCK ERP



Illustrative local app · actual browser capture

UP TO · EMPTY RUNNING

40%

DPIIT-NCAER, 2025

CAPACITY UTILISATION

74.6%

RBI OBICUS Q1 2026-27 [VERIFY]

VENTILATORS · ~3 MONTHS

~60,000

COVID sharing case [source VERIFY]

11 · SAP BTP

Built on *SAP BTP.*

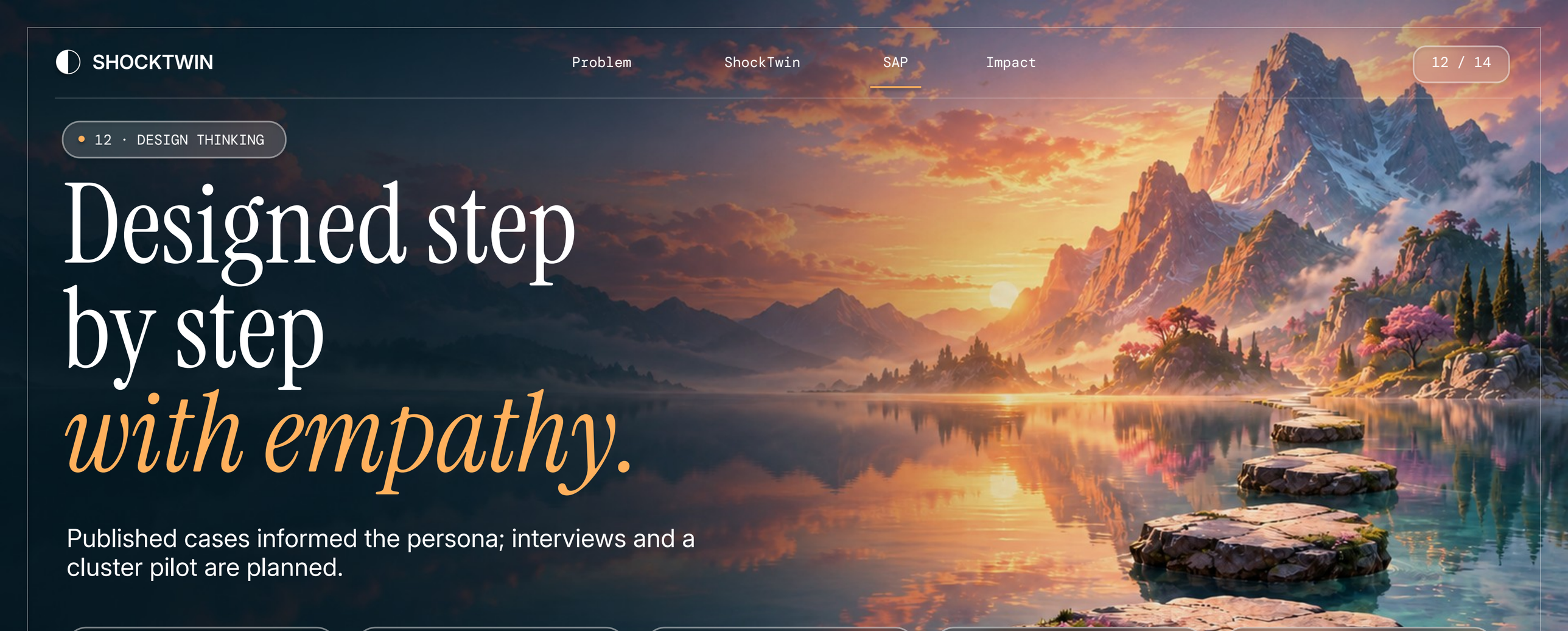
One decision workflow, with human approval at the boundary of action.

UI	SAPUI5 / Horizon	Project target/implementation
AGENT	Build Process Automation	SAP target; local fallback
APP	CAP / Node.js	Project target/implementation
AI	AI Core / Gen AI Hub	SAP target; local GraphSAGE/template fallback
DATA	HANA Cloud	SAP target; SQLite local
SOURCE	Integration Suite / S/4HANA	SAP target; mocks local

12 · DESIGN THINKING

Designed step by step *with empathy.*

Published cases informed the persona; interviews and a cluster pilot are planned.



EMPATHIZE

Published cases + persona

Project process record

DEFINE

Hidden dependency + response gap

Project process record

IDEATE

One connected loop

Project process record

PROTOTYPE

Working local app

Project process record

TEST

Engineering tests; interviews next

Project process record

13 · VALUE CONTEXT

What *resilience* is worth.

Published context motivates the product; these are not measured ShockTwin savings.

DECADE DISRUPTION COST

~45%

MGI 2020; fraction of one year's profits

LEAD TIME

-30.8%

Collaboration study from brief [original source VERIFY]

TRANSPORT COST

-25.83%

Collaboration study from brief [original source VERIFY]

MONTH-LONG SHOCK

~EVERY 3.7 YEARS

MGI, 2020

14 · NEXT STEP

Never be *surprised again.*

Next: a BTP-connected prototype and a pilot with one Bengaluru auto-parts cluster.

NEXT STEP

BTP + cluster pilot

Proposed project plan; no partner commitment

TEAM

Francis Reuben R · Avaneesh Affan · Vivek

P-ID

P2012988948

Team and P-ID: supplied brief